

Fig. 2: Configuration of bits

Communication. Tasks are good but tasks that can communicate with each other are even better. This brings us to the second FreeRTOS job, that is, communication. About 40 per cent of FreeRTOS's core codes deal with communication. queue.c and queue.h handle FreeRTOS communication.

Tasks and interrupts use queues to send data to each other and to signal the use of critical resources using semaphores and Mutexes.

Application of FreeRTOS running on an MCU

This article describes a FreeRTOS based application running on PIC18F4520 MCU. Three LEDs are connected on PORT B at pin numbers RD0, RD1 and RD2.

The scheduler will schedule tasks such as:

LED0 at pin RD0 - On for 1000ms

LED1 at pin RD1 - On for 2000ms

LED2 at pin RD2 - On for 3000ms

Delay is calculated using VTaskDelay () function which will cause a task to block for the specified num-

ber of ticks from the time VTaskDelay () is called.

System requirements are:

FreeRTOS v7.3.0

MPLAB IDE v8.89

MPLAB C18 compiler v3.45

MPLAB IC2D

PICDEM 2 Plus demonstration board

Steps to compile FreeRTOS on PIC18F4520

Creation of a project. Launch MPLAB IDE; go to Project→Open→ and type the name of your project (in this case, the name is LED Toggle .c). You may choose any name.

Click on Open. Untitled Workspace will appear on the screen as shown in Fig. 1.

Selection of the device and configuring the bits. Select Configure from Menu→Select Device 18F4520 from drop-down menu.

Again, go to Configure → Configuration bits. Uncheck box configuration bits set in code. Bits are to be selected as under:

Oscillator→XT

Disable all enable options

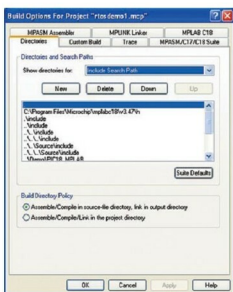


Fig. 3: Build options

Convert PORT B as a digital port using option PRADEN

Disable LVP

MCLR to be enabled option MCLRE

Then check Untick box

Finally, configuration bits of PIC18F4520 are configured as shown in Fig. 2.

Project configuration. Click Project→Build Options→Project. Specify include and library search path in Directories. Select Include Search Path from Show directories for combo box. Add the following five paths together:

```
./include
./mccl18/h
./FreeRTOS/Demo/Common/include
./FreeRTOS/Source/include
```

The screen shot of the above is shown in Fig. 3.

Library Search Path. Select Library Search Path from Show directories for combo box. Add C:\Program Files\Microchip\mplab18\lib directory from C18 compiler installation. The process is shown in the



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